

Supplementary Material

Segmentation Error and Boundary-Condition Tuning in Computed Coronary FFR: A Controlled In Silico Study

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S1. ANALYSIS PIPELINE

Fig. S1 shows the analysis pipeline. Host vessels were eligible with an image quality of at least 2 (0–4 scale), a taper-fit radius r_{fit} of at least 1.0 mm at the lesion center and native narrowing below 40% diameter stenosis (DS). The measurement point 20 mm beyond the lesion lay on a vessel of radius at least 0.75 mm, and the FFR before insertion was at least 0.90.

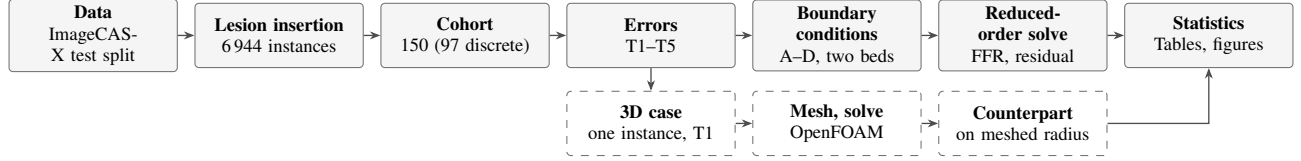


Fig. S1. Analysis pipeline. Shaded boxes: reduced-order arm, run for every instance in one automated run. Dashed boxes: the 3D case study. Sensitivity analyses and noise floors repeat the reduced-order arm.

S2. EXCLUSIONS

Table S1 lists every exclusion. Protocols C and D need at least two territories with a vessel. A Protocol C fit at its search bound is a failure (two half-voxel and eight ten-point throat-error models; 10 of 4940 floor draws). Protocol D fits at the $10^{\pm 3}$ bound keep a defined residual and are retained: two missed-branch models (residual 0.08 and 0.11) and 23 half-voxel and 41 ten-point throat-error models with an increased stenosis. Of these 64 throat-error models, 43 failed the check and 21 passed while materially wrong. Excluding them would raise throat passes-and-wrong by up to 13 points. Under Protocol A a vessel break can remove every outflow node (36 discrete, 2 leaky). Counted as reclassifications (their FFR is 1), these raise the discrete-bed vessel-break rate from 40% to 43%.

Table S1. Cells With Excluded Models

Bed	Error	Protocol	Solved	Excluded (reason: count)
Discrete	T1	all	77 of 97	no side branch distal to the lesion: 20
Discrete	T1	C	44	fewer than two territories with surviving vessels: 33
Discrete	T2	all	96 of 97	truncation point at a segment start: 1
Discrete	T2	A	60	no node with bed outflow remains: 36
Discrete	T2	C	60	fewer than two territories with surviving vessels: 36
Discrete	T5	C	193	Protocol C fit at search bound: 1
Leaky	T1	all	118 of 150	no side branch distal to the lesion: 32
Leaky	T1	C	71	fewer than two territories with surviving vessels: 47
Leaky	T2	all	149 of 150	truncation point at a segment start: 1
Leaky	T2	A	147	no node with bed outflow remains: 2
Leaky	T2	C	100	fewer than two territories with surviving vessels: 49
Leaky	T5	C	299	Protocol C fit at search bound: 1

S3. MIXED-EFFECTS MODEL OF RECLASSIFICATION

Table S2. Bayesian Logistic Model of Reclassification: Posterior Mean Log-Odds (95% Interval)

Term	Discrete bed	Leaky bed
Intercept	-2.63 (-2.85, -2.42)	-2.39 (-2.61, -2.18)
T2	0.69 (0.31, 1.07)	1.81 (1.47, 2.15)
T3	-1.92 (-2.40, -1.44)	-2.49 (-3.12, -1.87)
T4	-1.16 (-1.59, -0.73)	-1.00 (-1.45, -0.54)
B	-1.54 (-1.92, -1.17)	-1.76 (-2.27, -1.26)
C	-0.69 (-1.10, -0.28)	-0.99 (-1.42, -0.57)
T2 × B	0.06 (-0.55, 0.66)	-1.89 (-2.66, -1.12)
T3 × B	1.26 (0.41, 2.10)	1.13 (-0.06, 2.32)
T4 × B	0.72 (-0.08, 1.51)	-1.26 (-2.87, 0.35)
T2 × C	-0.73 (-1.46, 0.00)	-2.00 (-2.76, -1.24)
T3 × C	0.44 (-0.39, 1.28)	0.78 (-0.25, 1.81)
T4 × C	-0.10 (-0.89, 0.68)	0.21 (-0.54, 0.96)
Severity (% DS)	0.019 (0.016, 0.022)	0.019 (0.016, 0.022)
Length (mm)	-0.04 (-0.05, -0.02)	-0.08 (-0.09, -0.06)

Reference levels: T1 (missed branch) and Protocol A. Baseline-band terms are omitted from the table. Random intercepts for patient and lesion slot (vessel, position and length). The model agrees with the paired tests in direction. Variational inference; intervals are posterior mean ± 1.96 standard deviations.

S4. SENSITIVITY ANALYSES

A. Perfusion-Check Threshold

Table S3 gives topological passes-and-wrong at the 10% threshold and at 13% and 16%.

Table S3. Proportion of Models That Pass the Perfusion Check and Are Materially Wrong, % (Wilson 95% Interval), by Perfusion-Check Threshold

Bed	Errors	Protocol	<i>n</i>	Passes and wrong, % of <i>n</i>			At 10%	
				10%	13%	16%	Pass	Wrong among passing, %
Discrete	T1+T2	A	137	12 (8–19)	23 (17–31)	31 (24–40)	21	81 (60–92)
Discrete	T1+T2	B	137	1 (0–5)	11 (7–17)	20 (14–27)	23	9 (2–27)
Discrete	T1+T2	C	104	19 (13–28)	28 (20–37)	38 (30–48)	39	51 (36–66)
Discrete	T1+T2	D	104	20 (14–29)	20 (14–29)	20 (14–29)	103	20 (14–29)
Leaky	T1+T2	A	218	20 (15–26)	30 (25–37)	38 (31–44)	108	40 (31–49)
Leaky	T1+T2	B	218	5 (3–8)	5 (3–9)	6 (3–9)	176	6 (3–10)
Leaky	T1+T2	C	171	8 (5–13)	9 (6–15)	9 (6–15)	154	9 (5–15)
Leaky	T1+T2	D	171	5 (3–10)	5 (3–10)	5 (3–10)	171	5 (3–10)

Materially wrong: $|\Delta\text{FFR}| > 0.05$. *n* counts models with a defined perfusion residual: T2 under Protocols A and B in the leaky bed, *n* = 100; T2 under Protocol B in the discrete bed, *n* = 60. Protocol D matches every territory flow in all but two models, so its passes-and-wrong and wrong proportions differ by at most one model.

B. Hyperemic Demand

With $k = 562 \text{ s}^{-1}$, a vessel with the normal proximal LAD diameter of 3.7 mm carries 214 mL/min, within one standard deviation of the hyperemic flow measured in that artery by continuous thermodilution (228 ± 71 to 293 ± 102 mL/min). The cohort's median demand was 2.28 mL/s (137 mL/min) per tree. The demand constant *k* was scaled by 0.7 and 1.3 with the same cohort (Table S4). The ordering of topological against caliber errors, and of Protocols B and C against Protocol A, was unchanged in both beds. The magnitudes changed because demand moves the clean FFR (leaky-bed median 0.869, 0.801 and 0.741). Tuned topological passes-and-wrong fell to 3% (leaky bed, 0.7 times the demand) and stayed at 12–25% (discrete bed).

Table S4. Sensitivity to Hyperemic Demand

Bed	Demand scale	Reclass., % (Protocol A)		Topological reclass., %		Passes and wrong, % (C, topological)	Missed-branch ΔFFR median, A / C
		Topological	Caliber	B	C		
Discrete	0.7	32 (25–40)	6 (3–10)	12 (8–18)	18 (12–27)	12 (7–19)	0.070 / 0.057
	1.0	33 (26–41)	10 (7–15)	14 (10–20)	19 (13–28)	19 (13–28)	0.095 / 0.077
	1.3	40 (32–48)	5 (2–9)	16 (11–22)	19 (13–28)	25 (18–34)	0.106 / 0.091
Leaky	0.7	20 (16–26)	6 (4–9)	7 (4–10)	9 (5–14)	3 (1–7)	0.032 / 0.010
	1.0	32 (26–38)	6 (4–9)	5 (3–8)	8 (5–13)	8 (5–13)	0.047 / 0.015
	1.3	37 (32–43)	4 (2–7)	5 (3–9)	8 (4–13)	12 (8–18)	0.057 / 0.021

Demand scale 1.0 is the primary analysis ($k = 562 \text{ s}^{-1}$). Percentages with Wilson 95% intervals in parentheses. Missed-branch ΔFFR is over the instances in which Protocol C was defined.

C. Demand Replication With Reselected Cohorts

The whole pipeline was repeated with k doubled and tripled, with cohorts reselected at that demand. At three times the demand only 17 instances met the discrete-bed criteria, so the discrete arm is reported at twice the demand (Table S5). The excess of tuned over re-derived passes-and-wrong at twice the demand was significant in three of four caliber comparisons and in no topological one.

Table S5. Demand Replication: Reclassification and Passes-and-Wrong Rates, % (Wilson 95% Interval)

Bed	k scale	Inst.	Reclass., A		Topological reclass.		Topological passes and wrong			Caliber passes and wrong		
			Topol.	Caliber	C	D	B	C	D	B	C	D
Discrete	1	97	33 (26–41)	10 (7–15)	19 (13–28)	8 (4–14)	1 (0–5)	19 (13–28)	20 (14–29)	0 (0–2)	5 (2–9)	18 (13–24)
	2	47	50 (38–62)	1 (0–6)	20 (11–32)	8 (3–18)	6 (2–15)	14 (7–26)	20 (11–32)	0 (0–4)	1 (0–6)	14 (8–22)
Leaky	1	150	32 (26–38)	6 (4–9)	8 (5–13)	6 (3–10)	5 (3–8)	8 (5–13)	5 (3–10)	0 (0–1)	12 (9–17)	4 (2–7)
	2	150	34 (29–40)	5 (3–8)	7 (4–11)	3 (1–7)	2 (1–5)	5 (3–10)	3 (1–7)	0 (0–1)	13 (10–18)	5 (3–8)
	3	150	33 (28–39)	5 (3–8)	6 (4–11)	6 (4–11)	4 (2–8)	7 (4–12)	4 (2–8)	0 (0–2)	12 (9–16)	3 (1–5)

Scale 1 is the primary analysis ($k = 562 \text{ s}^{-1}$, primary cohort). Scales 2 and 3 use cohorts reselected at that demand. Passes and wrong: perfusion residual below 10% and $|\Delta\text{FFR}| > 0.05$; denominators are models with a defined residual, as in Table I of the main text.

D. Gray Zone

Beyond-zone reclassifications end outside 0.75–0.85 (Table S6). The zone held 30% (discrete) and 33% (leaky) of clean models.

Table S6. Reclassification, Overall and Beyond the Gray Zone (0.75–0.85), % of Models (Wilson 95% Interval)

Bed	Protocol	Topological errors			Caliber errors		
		n	Reclass.	Beyond zone	n	Reclass.	Beyond zone
Discrete	A	137	33 (26–41)	20 (15–28)	194	10 (7–15)	0 (0–2)
	B	173	14 (10–20)	5 (2–9)	194	8 (5–12)	0 (0–2)
	C	104	19 (13–28)	9 (5–16)	194	8 (5–12)	1 (0–3)
	D	104	8 (4–14)	2 (1–7)	194	11 (7–16)	1 (0–3)
Leaky	A	265	32 (26–38)	23 (18–28)	300	6 (4–9)	0 (0–1)
	B	267	5 (3–8)	0 (0–1)	300	1 (1–3)	0 (0–1)
	C	171	8 (5–13)	1 (0–3)	300	4 (2–7)	1 (0–2)
	D	171	6 (3–10)	0 (0–2)	300	4 (2–6)	0 (0–1)

S5. CALIBER ERROR AT THE STENOSIS THROAT

The throat error (T5) re-inserts each lesion with its throat diameter changed by half the in-plane voxel size (spacing median 0.352 mm; throat radius ± 0.088 mm; 7.3 points of DS, 4.5–9.8) or its DS changed by ± 10 points. The ± 10 -point magnitude lies within the published CT-versus-invasive disagreement (standard deviation 12.3–12.4 points). Protocol C fits at the search bound were excluded (2 at half a voxel, 8 at ± 10 points). Bound fits under Protocol D are described in Section S2.

Table S7. Throat Caliber Error: Reclassification and Passes-and-Wrong, % of Models (Wilson 95% Interval)

Error	Protocol	Discrete bed (97 instances)				Leaky bed (150 instances)			
		<i>n</i>	Reclass.	Passes and wrong	Beyond zone	<i>n</i>	Reclass.	Passes and wrong	Beyond zone
DS ± 10 pooled	A, B	194	40 (33–47)	20 (15–26)	27 (22–34)	300	42 (37–48)	40 (34–45)	35 (30–40)
	C	187	44 (37–52)	51 (44–58)	37 (30–44)	299	43 (38–49)	62 (56–67)	41 (35–46)
	D	194	45 (38–52)	75 (68–80)	40 (34–47)	300	44 (38–49)	86 (81–89)	40 (35–46)
$\pm \frac{1}{2}$ voxel, $k \times 2$	A, B	94	27 (19–36)	35 (26–45)	12 (7–20)	300	29 (24–35)	44 (39–50)	12 (9–16)
	C	94	31 (22–41)	52 (42–62)	13 (7–21)	300	35 (30–40)	69 (64–74)	17 (13–21)
	D	94	34 (25–44)	73 (64–81)	18 (12–27)	300	35 (30–41)	75 (70–80)	18 (14–22)
$\pm \frac{1}{2}$ voxel, $k \times 3$	A, B	–	–	–	–	300	29 (24–34)	48 (43–54)	9 (6–13)
	C	–	–	–	–	300	34 (29–40)	68 (62–73)	15 (12–20)
	D	–	–	–	–	300	33 (28–39)	72 (67–77)	14 (10–18)

$\pm \frac{1}{2}$ voxel: throat diameter narrowed or widened by half the in-plane voxel size, both signs pooled. DS ± 10 : diameter stenosis raised or lowered by 10 points, both signs pooled. Passes and wrong: perfusion residual below 10% and $|\Delta FFR| > 0.05$. Beyond zone: reclassifications whose corrupted FFR lies outside 0.75–0.85. Half-voxel rows at the primary demand are in Table I of the main text. Their beyond-zone proportions were 14%, 22% and 26% (Protocols A and B, C, D; discrete) and 21%, 29% and 29% (leaky). Protocols A and B coincide for throat errors. Protocol C fits on the search bound are excluded, which reduces n . $k \times 2$, $k \times 3$: demand constant doubled or tripled with the cohort reselected at that demand. Discrete bed: 47 instances at $k \times 2$, not analyzed at $k \times 3$. Topological reclassification under Protocol A was 50% (38–62%, discrete) and 34% (29–40%, leaky) at $k \times 2$ and 33% (28–39%, leaky) at $k \times 3$. At the primary demand it was 33% (26–41%) and 32% (26–38%), and caliber errors away from the throat gave 10% (7–15%) and 6% (4–9%).

S6. TERRITORY GRANULARITY

Protocols C and D were repeated with each main-branch territory divided again at its first bifurcation (median four territories, discrete; five, leaky). A territory without a surviving vessel entered the residual as a 100% mismatch. Without this rule, discrete topological passes-and-wrong would have risen to 26%. With the main-branch partition the procedure reproduced the original results exactly (Table S8). Throat-error models that left the passes-and-wrong count all failed the finer check. None had its FFR corrected, and 76–88% of tuned throat-error models that passed remained materially wrong. Reclassifications changed by at most five per cell. The simulated noise floor was computed at main-branch level only.

Table S8. Passes-and-Wrong, % (Wilson 95% Interval), at Main-Branch and Finer Territory Level

Bed	Errors	Protocol	n	Main branch	Finer	p
Discrete	T1+T2	C	104	19 (13–28)	7 (3–13)	< 0.001
		D	104	20 (14–29)	3 (1–8)	< 0.001
	T3+T4	C	194	5 (2–9)	5 (3–9)	1.0
		D	194	18 (13–24)	23 (18–30)	0.002
	T5, $\pm\frac{1}{2}$ voxel	A, B	194	29 (23–36)	25 (20–32)	0.016
		C	193	57 (50–64)	51 (44–58)	0.003
		D	194	80 (74–85)	78 (71–83)	0.12
Leaky	T1+T2	C	171	8 (5–13)	5 (2–9)	0.07
		D	171	5 (3–10)	2 (1–6)	0.18
	T3+T4	C	300	12 (9–17)	12 (9–17)	1.0
		D	300	4 (2–7)	9 (7–13)	< 0.001
	T5, $\pm\frac{1}{2}$ voxel	A, B	300	42 (37–48)	35 (30–41)	< 0.001
		C	299	69 (63–74)	60 (55–66)	< 0.001
		D	300	81 (76–85)	81 (76–85)	1.0

Passes and wrong: perfusion residual below 10% and $|\Delta FFR| > 0.05$. Main branch: subtrees below each child of the first bifurcation (two or three per tree). Finer: each divided again at its own first bifurcation (median four, discrete; five, leaky). n counts models with a defined residual at both levels; Protocol C fits on the search bound at either level are excluded. For throat errors the FFR of Protocols A and B is the same at both levels and only the residual changes. p : exact McNemar test, paired by model.

S7. OVERLAP AND TOPOLOGY OF THE CORRUPTED TREES

Each tree was represented as cylinders of node radius and element length (a tube model) and compared with its clean tree. No error changed the number of connected components (Table S9). Among the topological errors that changed the decision, 91% (79–96%, discrete) and 62% (51–72%, leaky) kept a whole-scan DSC at or above the inter-observer 0.928. The AUC of tree DSC for a Protocol A reclassification was 0.65 (discrete) and 0.79 (leaky) over T1–T4 and 0.54 and 0.70 within topological errors. The taper DSC exceeds 0.928 because it narrows only the tree beyond the lesion (about 40% of its volume).

Table S9. Tube-Model Overlap of Corrupted Trees, Median (IQR), and Protocol A Decision Changes

Bed	Error	<i>n</i>	DSC, tree	DSC, scan	clDice	Flow lost, %	Reclass. A, %
Discrete	T1	77	0.975 (0.947–0.984)	0.985 (0.977–0.991)	0.940 (0.903–0.963)	27 (14–43)	27 (19–38)
	T2	96	0.893 (0.701–0.937)	0.935 (0.888–0.960)	0.847 (0.544–0.907)	30 (15–100)	40 (29–53) ^a
	T3	97	0.996 (0.995–0.997)	0.998 (0.998–0.998)	1.000	0	7 (4–14)
	T4	97	0.971 (0.947–0.981)	0.983 (0.974–0.989)	1.000	0	13 (8–22)
	T5	194	0.998 (0.998–0.999)	0.999 (0.999–0.999)	1.000	0	30 (24–37)
Leaky	T1	118	0.969 (0.946–0.980)	0.983 (0.975–0.990)	0.930 (0.899–0.957)	9 (6–17)	18 (12–26)
	T2	149	0.895 (0.721–0.936)	0.934 (0.886–0.962)	0.855 (0.517–0.899)	14 (7–56)	43 (35–51)
	T3	150	0.997 (0.995–0.998)	0.998 (0.998–0.999)	1.000	0	2 (1–6)
	T4	150	0.972 (0.951–0.981)	0.983 (0.976–0.988)	1.000	0	9 (6–15)
	T5	300	0.999 (0.998–0.999)	0.999 (0.999–0.999)	1.000	0	36 (31–42)
T2, right coronary artery			0.62–0.63	0.85–0.86	0.45–0.49		

Reclass. A: under Protocol A, Wilson 95% interval. T5: both half-voxel throat errors pooled. Flow lost: share of the clean model's bed outflow on deleted nodes. ^a *n* = 60; right coronary breaks leave no discrete outlet under Protocol A.

S8. DETECTION BEFORE AND DURING TUNING

Positives are error models, and negatives are the correct-anatomy draws of the simulated floor before tuning. Correct anatomy without noise has a residual below 10^{-8} . With noise its median is 0.10, and 48% (discrete) and 50% (leaky) of draws exceed the 10% check. With noise-free targets the pre-tuning residual of topological errors gave an AUC of 0.77 (0.71–0.82, discrete) and 0.22 (0.18–0.27, leaky) and flagged 83% and 19% at the 10% check. With the error models' targets carrying the same noise (Table S10), the AUC was 0.82 (discrete) and 0.56 (leaky).

S9. NOISE FLOORS

The perfusion-check bound is $1.96 \times (10-15\%) \approx 20-29\%$ for one model output against one measurement, from the within-subject variation of hyperemic myocardial blood flow. The simulated floor perturbs, in 20 draws per instance, cardiac output (relative standard deviation 0.05), aortic pressure (0.056), viscosity (0.02), territory flow shares (0.10) and the measured territory flows (0.083). Tuned as in Protocol C, correct anatomy was reclassified in 6.2% (discrete) and 7.2% (leaky) of draws (median $|\Delta\text{FFR}|$ 0.012, 95th percentile 0.05) and passed the check in 72%. For Table S10, every error model was tuned to the floor's noisy targets, and a Protocol D floor was added. Rates are means over 20 draws per instance with patient-level cluster-bootstrap intervals. The excess is the per-draw difference from the correct-anatomy model of the same instance, bed and draw, and *n* counts models with a defined residual.

Table S10. Passes-and-Wrong, % of Models, With Error Models and Correct Anatomy Tuned to the Same Noisy Targets (Noise-Free: Table I)

Error	Protocol	Discrete bed				Leaky bed			
		<i>n</i>	Noise-free	Matched	Excess over floor	<i>n</i>	Noise-free	Matched	Excess over floor
Topological (T1+T2)	C	104	19	19 (14–25)	17 (12–22)	171	8	8 (5–11)	7 (4–9)
	D	104	20	36 (28–44)	16 (9–23)	171	5	19 (15–23)	6 (3–8)
Caliber (T3+T4)	C	194	5	9 (7–11)	5 (4–7)	300	12	12 (10–15)	8 (6–10)
	D	194	18	28 (24–32)	13 (10–16)	300	4	13 (11–16)	4 (3–6)
Throat ($\pm \frac{1}{2}$ voxel)	C	194	57	42 (37–47)	39 (35–44)	300	69	49 (45–53)	45 (41–49)
	D	194	80	78 (72–85)	63 (56–70)	300	81	80 (75–85)	71 (66–76)
Correct anatomy	C	97	0	3 (2–4)	–	150	0	4 (2–6)	–
	D	97	0	15 (12–19)	–	150	0	9 (7–11)	–

S10. THREE-DIMENSIONAL CASE STUDY: MESH CHECKS

Surfaces were extracted by marching cubes with Taubin smoothing. Meshes of 3.5–3.9 million cells (cfMesh) had 25 μm cells within ± 4 mm of the throat and four boundary layers. OpenFOAM simpleFoam (ESI v2406; SIMPLE; bounded second-order upwind) ran 3 000 iterations per solve to scaled residuals below 10^{-5} . A finer throat zone (12.5 μm , 6.24 and 8.31 million cells) raised the baseline FFR from 0.8698 to 0.8716 and 0.8718 (0.0019–0.0021). Outlet flows changed by at most 0.25%, above the idealized-case discretization uncertainty of 5.5×10^{-4} , so the patient-case uncertainty is about 0.002, against an error effect of 0.074. The finest mesh missed the strict residual limit (velocity residual 1.08×10^{-5}) although the monitored inlet pressure was steady. The lesion carried about 10 mL/min (tree inflow 0.68 mL/s in 3D, 0.65–0.67 mL/s in the cohort model). At the meshed throat radius of 0.276 mm the mean velocity is about 0.7 m/s and the Reynolds number about 110, within the laminar regime.